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MAPT ANTISENSE OLIGONUCLEOTIDE

Abstract

Provided herein are MAPT antisense oligonucleotides and compositions comprising a MAPT antisense oligonucleotide. Also provided herein are methods of using the MAPT antisense oligonucleotides or compositions comprising a MAPT antisense oligonucleotide for reducing MAPT expression and/or treating tauopathy in a subject.

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Background/Summary

SEQUENCE LISTING

[0001] The present application is being filed along with a Sequence Listing in ST.26 XML format. The Sequence Listing is provided as a file titled “31012_US” created Feb. 6, 2025, and is 726 kilobytes in size. The Sequence Listing information in the ST.26 XML format is incorporated herein by reference in its entirety.

BACKGROUND

[0002] Microtubule associated protein Tau (MAPT) is encoded by the MAPT gene located on chromosome 17. Tau protein interacts with tubulin to stabilize the microtubules and promote tubulin assembly into microtubules. MAPT transcripts are differentially expressed throughout the body, predominantly in the central and peripheral nervous system.

[0003] The MAPT gene consists of 16 exons. Alternative mRNA splicing gives rise to multiple MAPT isoforms. At least six Tau isoforms exist in human brain, ranging from 352 to 441 amino acids long. Alternative splicing of exons 2 and/or 3 leads to inclusion of zero, one, or two copies of the N-terminal acidic domain, which are referred to as 0N, 1N, or 2N Tau, respectively. The Tau isoforms that include exon 10, which encodes an additional microtubule-binding domain, are referred to as “4R Tau”, as it has four microtubule-binding domains. The Tau isoforms without exon 10 are referred to as “3R Tau”, as it has three microtubule-binding domains.

[0004] Mutations in MAPT and hyperphosphorylation of Tau protein can cause aggregation and deposition of Tau in pathogenic neurofibrillary tangles, causing progressive neurodegenerative disorders such as Alzheimer's disease, frontotemporal dementia (FTD), progressive supranuclear palsy (PSP), and other tauopathies.

[0005] Antisense oligonucleotides (ASOs) are powerful and versatile agents being used in a number of applications including mRNA reduction, translation arrest, miRNA inhibition, splicing modulation, and polyadenylation site selection. MAPT antisense oligonucleotides have been described before, e.g., in US 2016/0145617.

[0006] There is currently no FDA-approved disease-modifying therapeutic agent specifically for reducing MAPT and treating tauopathies. Accordingly, there remains a need for therapeutic agents that can reduce or adjust MAPT expression for treating tauopathies, e.g., by utilizing antisense oligonucleotides.

SUMMARY OF INVENTION

[0007] Provided herein are MAPT antisense oligonucleotides and compositions comprising a MAPT antisense oligonucleotide. Also provided herein are methods of using the MAPT antisense oligonucleotides or compositions comprising a MAPT antisense oligonucleotide for reducing MAPT expression and/or treating tauopathy in a subject.

[0008] In one aspect, provided herein are antisense oligonucleotides targeting MAPT. In some embodiments, provided herein are MAPT antisense oligonucleotides that comprises a nucleic acid sequence selected from any one of SEQ ID NOs: 1-11, 30-32, or 36-63, wherein optionally one or more nucleotides are independently modified nucleotides, and wherein optionally one or more internucleotide linkages are modified internucleotide linkages. In some embodiments, the antisense oligonucleotide is single stranded.

[0009] The MAPT antisense oligonucleotides described herein may include modifications. The modifications can be made to one or more nucleotides (e.g., to nucleobase, sugar, or phosphate) or to one or more internucleotide linkages.

[0010] In some embodiments, the antisense oligonucleotide provided herein comprises one or more modified nucleotides. In some embodiments, such modified nucleotides comprise a modified nucleobase, e.g., 5-methylcytosine. In some embodiments, each C in the antisense oligonucleotide

is a 5-methylcytosine. In some embodiments, such modified nucleotides comprise a modified sugar, e.g., a modified ribose or deoxyribose. In some embodiments, the modified sugar is a 2'-O-methoxyethyl (2'-O-MOE) modified sugar, 2'-O-methyl modified sugar, or 2'-fluoro modified sugar. In some embodiments, the modified sugar is a 2'-O-MOE modified sugar. In some embodiments, the modified sugar is a locked nucleic acid (LNA).

[0011] In some embodiments, the antisense oligonucleotides provided herein are 12 to 30 nucleotides in length. In some embodiments, the MAPT antisense oligonucleotides are 15 to 25 nucleotides in length. In some embodiments, the MAPT antisense oligonucleotides are 16 to 20 nucleotides in length. In some embodiments, the MAPT antisense oligonucleotides are 18 to 20 nucleotides in length. In some embodiments, the MAPT antisense oligonucleotides are 20 nucleotides in length. In some embodiments, the MAPT antisense oligonucleotides are 16 nucleotides in length.

[0012] In some embodiments, the MAPT antisense oligonucleotide provided herein are gapmers, wherein the first to fifth nucleotides each comprise a 2'-O-MOE modified sugar, wherein the sixth to fifteenth nucleotides each comprise a 2'-deoxynucleoside, and wherein the sixteenth to twentieth nucleotides each comprise a 2'-O-MOE modified sugar.

[0013] In some embodiments, the MAPT antisense oligonucleotide provided herein are gapmers, wherein the first to third nucleotides each comprise LNA, wherein the fourth to thirteenth nucleotides each comprise a 2'-deoxynucleoside, and wherein the fourteenth to sixteenth nucleotides each comprise LNA.

[0014] In some embodiments, the antisense oligonucleotides provided herein comprise one or more modified internucleotide linkages. In some embodiments, the modified internucleotide linkage is a phosphorothioate linkage. In some embodiments, the antisense oligonucleotides provided herein comprise phosphodiester and phosphorothioate internucleotide linkage. In some embodiments, the internucleotide linkages of the MAPT antisense oligonucleotides are sooooooooooooo, or sooooooooooooooooo, from 5' end to 3' end, wherein each s is a phosphorothioate linkage and each o is a phosphodiester linkage. In some embodiments, the internucleotide linkages of the MAPT antisense oligonucleotides are all phosphorothioate linkages.

[0015] In some embodiments, the antisense oligonucleotides provided herein comprise a sequence selected from any one of SEQ ID NOs: 12-22, 33-35, or 64-94. In some embodiments, the antisense oligonucleotides provided herein consist of a sequence selected from any one of SEQ ID NOs: 12-22, 33-35, or 64-94.

[0016] In another aspect, provided herein are pharmaceutical compositions comprising a MAPT antisense oligonucleotide described herein and a pharmaceutically acceptable carrier. Also provided herein are pharmaceutical compositions comprising a means for reducing MAPT expression in a cell and a pharmaceutically acceptable carrier.

[0017] In another aspect, provided herein are methods of reducing MAPT expression in a patient in need thereof, and such method comprises administering to the patient an effective amount of a MAPT antisense oligonucleotide or a pharmaceutical composition described herein. The MAPT antisense oligonucleotide can be administered to the patient intrathecally, intravenously, subcutaneously, or via intracisternal magna injection.

[0018] In another aspect, provided herein are methods of treating a tauopathy in a patient in need thereof, and such method comprises administering to the patient an effective amount of the MAPT antisense oligonucleotide or a pharmaceutical composition described herein.

[0019] Also provided herein are methods of reducing MAPT expression in a cell (e.g., a neuron), and such methods can include contacting the cell with a MAPT antisense oligonucleotide described herein; and incubating the cell for a time sufficient for degradation of MAPT mRNA, thereby reducing MAPT expression in the cell.

[0020] In another aspect, provided herein are MAPT antisense oligonucleotides or pharmaceutical compositions comprising a MAPT antisense oligonucleotide for use in reducing MAPT expression.

Also provided herein are MAPT antisense oligonucleotides or the pharmaceutical composition comprising a MAPT antisense oligonucleotide for use in a therapy. Also provided herein are MAPT antisense oligonucleotides or pharmaceutical compositions comprising a MAPT antisense oligonucleotide for use in the treatment of a tauopathy. Also provided herein are uses of MAPT antisense oligonucleotides in the manufacture of a medicament for the treatment of a tauopathy.

Description

DETAILED DESCRIPTION

[0021] Provided herein are MAPT antisense oligonucleotides and compositions comprising a MAPT antisense oligonucleotide. Also provided herein are methods of using the MAPT antisense oligonucleotides or compositions comprising a MAPT antisense oligonucleotide for reducing MAPT expression and/or treating tauopathy in a subject.

[0022] An antisense oligonucleotide usually comprises a nucleic acid sequence complementary to the sequence of a target nucleic acid, e.g., a target genomic sequence, pre-mRNA, or mRNA sequence. Hybridization occurs when hydrogen bonding (e.g., Watson-Crick, Hoogsteen or reversed Hoogsteen hydrogen bonding) forms between the complementary nucleobases of the antisense oligonucleotide and the target nucleic acid. Non-complementary nucleobases between an antisense oligonucleotide and a target nucleic acid may be tolerated provided that the antisense oligonucleotide remains able to specifically hybridize to a target nucleic acid.

[0023] Antisense oligonucleotides can be designed to modulate the expression of a target protein through either RNase H-dependent or RNase H-independent manners (see Watts J K, et al., J Pathol. 2012 January; 226(2): 365-379). When an antisense oligonucleotide comprising a contiguous stretch of DNA hybridizes with a target RNA, the DNA-RNA heteroduplex recruit RNase H, which cleaves the target RNA in the duplex and promotes subsequent degradation of the RNA fragments by cellular nucleases. An antisense oligonucleotide can also decrease target expression independent of RNase H by sterically blocking pre-mRNA processing or translation of mRNA into protein or modulating splicing of pre-mRNA into mature mRNA.

[0024] In one aspect, provided herein are antisense oligonucleotides targeting MAPT. In some embodiments, exemplary nucleic acid sequences of MAPT antisense oligonucleotides are provided in Table 1. In some embodiments, provided herein are MAPT antisense oligonucleotides that comprises a nucleic acid sequence selected from any one of SEQ ID NOs: 1-11, 30-32, or 36-63, wherein optionally one or more nucleotides are independently modified nucleotides, and wherein optionally one or more internucleotide linkages are modified internucleotide linkages. In some embodiments, the antisense oligonucleotide is single stranded. In some embodiments, the antisense oligonucleotides provided herein consist of a sequence selected from any one of SEQ ID NOs: 1-11, 30-32, or 36-63.

TABLE-US-00001 TABLE 1 Nucleic Acid Sequences of MAPT Antisense Oligonucleotides (ASO) Target region of human MAPT sequence NG_007398.2 MAPT									
SEQ	SEQ (range	ASO ID	ID 5092-138872)	No. Unmodified	Sequence NO	Modified			
Sequence NO (SEQ ID NO: 23)	1	CTTTCTTGGAAAGTATATTC	1						
[eCm]seToeToeTo[eCm]sd	12	72350-72369	TsdTsdGsdGsdAsdAsdAsd						
GsdTsdAseToeAoeTseTs [eCm]	2	GCTCAGAGCTACAGCAAGTC	2						
eGs[eCm]oeTo[eCm]oeAs	13	82367-82386	dGsdAsdGs[dCm]sdTsdAs						
[dCm]sdAsdGs[dCm]seAo eAoeGseTs[eCm]	3	GCTGCAGGGAAAGAGTTCTT	3						
eGs[eCm]oeToeGo[eCm]s	14	22592-22611	dAsdGsdGsdGsdAsdAsdA						
sdGsdAsdGseToeTo[eCm] seTseT	4	GATGGAGACAGTGTGTATCT	4						
eGseAoeToeGoeGsdAsdG	15	19124-19143	sdAs[dCm]sdAsdGsdTsdG						
sdTsdGseToeAoeTs[eCm] seT	5	GCAGATGGGGCCCAAATCCT	5						

eGs[eCm]oeAoeGsdTs 16 74462-74481 dGsdGsdGs[dCm]s
[dCm]s[dCm]sdAsdAseAoeTo [eCm]s[eCm]seT 6 GCCACAGTGACAGCAAATTC 6
eGs[eCm]o[eCm]oeAo 17 36467-36486 [eCm]sdAsdGsdTsdGsdAs
[dCm]sdAsdGs[dCm]sdAseAo eAoeTseTs[eCm] 7 GCCACAGACTCGAGTATCAG 7
eGs[eCm]o[eCm]oeAo 18 45989-46008 [eCm]sdAsdGsdAs[dCm]sdTs
[dCm]sdGsdAsdGsdTseAo eTo[eCm]seAseG 8 CCAAACGTGACTTGGCTCCT 8
[eCm]s[eCm]oeAoeAoeAs 19 105626-105645 [dCm]sdGsdTsdGsdAs
[dCm]sdTsdTsdGsdGs[eCm]o eTo[eCm]s[eCm]seT 9 TATGGACACACATGGGTGTA 9
eTseAoeToeGoeGsdAs 20 30079-30098 [dCm]sdAs[dCm]sdAs[dCm]s
dAsdTsdGsdGseGoeToeG seTseA 10 ACTAAAGACGTGTCTTTTGG 10
eAs[eCm]oeToeAoeAsdAs 21 13464-13483 dGsdAs[dCm]sdGsdTsdGs
dTsd[dCm]sdTseToeToeTse GseG 11 GTCATCTCTAGGTGACTTCC 11 eGseTo[eCm]oeAoeTs[dC
22 79347-79366 m]sdTs[dCm]sdTsdAsdGs dGsdTsdGsdAs[eCm]oeTo eTs[eCm]s[eCm] 12
GCCTTTGCGAGCGTGCACAG 30 eGs[eCm]o[eCm]oeToeTs 33 898-917
dTsdGs[dCm]sdGsdAsdGs [dCm]sdGsdTsdGs[eCm]o eAo[eCm]seAseG 13
TTTCTTGTGTTGGCAACTGGG 31 eTseToeTo[eCm]oeTsdTsd 34 80673-80692
GsdTsdTsdGsdGsdGs[dCm] sdAsdAs[eCm]oeToeGs eGseG 14 CTGGTGTGGGTGTTATTCCT
32 [eCm]seToeGoeGoeTsdGs 35 88009-88028 dTsdGsdGsdGsdTsdGsdTs
dTsdAseToeTo[eCm]s [eCm]seT 15 GTGGATCTAGTTCTCATCCA 36 eGseToeGseGseAsdTsdC
64 75985-76004 msdTsdAsdGsdTsdTsdCm sdTsdCmseAseTo[eCm]s [eCm]seA 16
GGATAGAATCCAGTTAATTA 37 eGseGoeAseTseAsdGsdAs 65 42171-42190
dAsdTsdCmsdCmsdAsdGs dTsdTseAseAoeTseTseA 17 TGTCACATCTAGAAACCACC 38
eTseGoeTs[eCm]seAsdCm 66 79820-79839 sdAsdTsdCmsdTsdAsdGsd
AsdAsdAs[eCm]s[eCm]oe As[eCm]s[eCm] 18 GCCTGGACAGAGGGATGAGC 39
eGs[eCm]o[eCm]seTseGsd 67 13357-13376 GsdAsdCmsdAsdGsdAsd
GsdGsdGsdAseTseGoeAse Gs[eCm] 19 AAGCTCCAGGTGATTGAGAC 40
eAseAoeGs[eCm]seTsdC 68 124341-124360 msdCmsdAsdGsdGsdTsdG
sdAsdTsdTseGseAoeGseA s[eCm] 20 TTCTGATACATTGTCAAAGT 41
eTseTo[eCm]seTseGsdAsd 69 10260-10279 TsdAsdCmsdAsdTsdTsdG
sdTsdCmseAseAoeAseGse T 21 AGCAGCTTGAAGGAGCCGGC 42 eAseGo[eCm]seAseGsdC
70 132887-132906 msdTsdTsdGsdAsdAsdGs dGsdAsdGs[eCm]s[eCm]o eGseGs[eCm] 22
TGGCTGATGATACCATTTGG 43 eTseGoeGs[eCm]seTsdGs 71 39953-39972
dAsdTsdGsdAsdTsdAsdC msdCmsdAseTseToeTseG seG 23 CTTTCTTGGAAGTAT 44
[CmL]s[TL]s[TL]sdTsdC 72 72354-72369 msdTsdTsdGsdGsdAsdAs dAsdGs[TL]s[AL]s[TL]
24 TTTCTTGGAAGTATA 45 [TL]s[TL]s[TL]sdCmsdTsd 73 72353-72368
dTsdGsdGsdAsdAsdAsdG sdTs[AL]s[TL]s[AL] 25 TTCTTGGAAGTATAT 46
[TL]s[TL]s[CmL]sdTsdTs 74 72352-72367 dGsdGsdAsdAsdAsdGsdT sdAs[TL]s[AL]s[TL] 26
TCTTGGAAGTATATT 47 [TL]s[CmL]s[TL]sdTsdGs 75 72351-72366
dGsdAsdAsdAsdGsdTsdA sdTs[AL]s[TL]s[TL] 27 CTTGGAAAGTATATTC 48
[CmL]s[TL]s[TL]sdGsdGs 76 72350-72365 dAsdAsdAsdGsdTsdAsdTsd dAs[TL]s[TL]s[CmL]
28 GTGGATCTAGTTCTCA 49 [GL]s[TL]s[GL]sdGsdAsd 77 75989-76004
TsdCmsdTsdAsdGsdTsdTs dCms[TL]s[CmL]s[AL] 29 TGGATCTAGTTCTCAT 50
[TL]s[GL]s[GL]sdAsdTsd 78 75988-76003 CmsdTsdAsdGsdTsdTsdC msdTsd[CmL]s[AL]s[TL]
30 GGATCTAGTTCTCATC 51 [GL]s[GL]s[AL]sdTsdCms 79 75987-76002
dTsdAsdGsdTsdTsdCmsd TsdCms[AL]s[TL]s[CmL] 31 GATCTAGTTCTCATCC 52
[GL]s[AL]s[TL]sdCmsdTsd 80 75986-76001 dAsdGsdTsdTsdCmsdTsd CmsdAs[TL]s[CmL]s
[CmL] 32 ATCTAGTTCTCATCCA 53 [AL]s[TL]s[CmL]sdTsdAs 81 75985-76000
dGsdTsdTsdCmsdTsdCms dAsdTsd[CmL]s[CmL]s[AL] 33 TTCTGATACATTGTCA 54
[TL]s[TL]s[CmL]sdTsdGs 82 10264-10279 dAsdTsdAsdCmsdAsdTsd TsdGs[TL]s[CmL]s[AL]
34 TCTGATACATTGTCAA 55 [TL]s[CmL]s[TL]sdGsdAs 83 10263-10278

dTsdAsdCmsdAsdTsdGsdTs[CmL]s[AL]s[AL] 35 CTGATACATTGTCAA 56
 [CmL]s[TL]s[GL]sdAsdTs 84 10262-10277 dAsdCmsdAsdTsdTsdGsd TsdCms[AL]s[AL]s[AL]
 36 TGATACATTGTCAAAG 57 [TL]s[GL]s[AL]sdTsdAsd 85 10261-10276
 CmsdAsdTsdTsdGsdTsdC msdAs[AL]s[AL]s[GL] 37 GATACATTGTCAAAGT 58
 [GL]s[AL]s[TL]sdAsdCms 86 10260-10275 dAsdTsdTsdGsdTsdCmsd AsdAs[AL]s[GL]s[TL]
 38 AGCAGCTTGAAGGAGC 59 [AL]s[GL]s[CmL]sdAsdG 87 132891-132906
 sdCmsdTsdTsdGsdAsdAsd GsdGs[AL]s[GL]s[CmL] 39 GCAGCTTGAAGGAGCC 60
 [GL]s[CmL]s[AL]sdGsdC 88 132890-132905 msdTsdTsdGsdAsdAsdGs
 dGsdAs[GL]s[CmL]s[CmL] 40 CAGCTTGAAGGAGCCG 61 [CmL]s[AL]s[GL]sdCmsd 89
 132889-132904 TsdTsdGsdAsdAsdGsdGsd AsdGs[CmL]s[CmL]s[GL] 41
 AGCTTGAAGGAGCCGG 62 [AL]s[GL]s[CmL]sdTsdTs 90 132888-132903
 dGsdAsdAsdGsdGsdAsdG sdCms[CmL]s[GL]s[GL] 42 GCTTGAAGGAGCCGGC 63
 [GL]s[CmL]s[TL]sdTsdGs 91 132887-132902 dAsdAsdGsdGsdAsdGsdC
 msdCms[GL]s[GL]s[CmL] 43 CTTTCTTGGAAGTATATTC 1 [eCm]seToeTseTs[eCm]sd 92
 72350-72369 TsdTsdGsdGsdAsdAsdAsd GsdTsdAseTseAoeTseTs [eCm] 44
 GCTGCAGGGAAAGAGTTCTT 3 eGs[eCm]oeTseGs[eCm]sd 93 22592-22611
 AsdGsdGsdGsdAsdAsdAs dGsdAsdGseTseTo[eCm]s eTseT 45 GATGGAGACAGTGTGTATCT 4
 eGseAoeTseGseGsdAsdGs 94 19124-19143 dAs[dCm]sdAsdGsdTsdGs

dTsdGseTseAoeTs[eCm]se T Abbreviations-“Cm” stands for 5-methylcytidine; “d” indicates 2'-deoxynucleotides; “e” indicates a 2'-O-MOE modified nucleotide; “L indicates a LNA modified nucleotide”; “s” indicates phosphorothioate linkage; “o” indicates phosphodiester linkage.

[0025] In some embodiments, the MAPT antisense oligonucleotide described herein can comprise a sequence that has 1, 2, or 3 differences from the sequences listed in Table 1. Such MAPT antisense oligonucleotides may still be capable of binding to the target nucleic acid, with 1, 2, or 3 mismatches, when a sufficient number of nucleobases of the antisense oligonucleotide form hydrogen bond with the corresponding nucleobases of the target nucleic acid.

[0026] In some embodiments, the antisense oligonucleotides provided herein are 12 to 30 nucleotides in length. In some embodiments, the MAPT antisense oligonucleotides are 15 to 25 nucleotides in length. In some embodiments, the MAPT antisense oligonucleotides are 16 to 20 nucleotides in length. In some embodiments, the MAPT antisense oligonucleotides are 18 to 20 nucleotides in length. In some embodiments, the MAPT antisense oligonucleotides are 20 nucleotides in length. In some embodiments, the MAPT antisense oligonucleotides are 16 nucleotides in length.

[0027] The MAPT antisense oligonucleotides described herein may include modifications. The modifications can be made to one or more nucleotides (e.g., to nucleobase, sugar, or phosphate) or to one or more internucleotide linkages, which are the bonds between two nucleotides.

Modifications can improve stability, efficacy, and/or reduce immunogenicity of the antisense oligonucleotides. For example, antisense oligonucleotides can be modified to have increased resistance to nucleases, increased stability, increased half-life, enhanced binding affinity for nucleic acid target, enhanced cellular uptake, and/or increased inhibitory activity when compared to the unmodified oligonucleotides.

[0028] In some embodiments, the antisense oligonucleotide provided herein comprises one or more modified nucleotides. In some embodiments, such modified nucleotides comprise a modified nucleobase, e.g., 5-methylcytosine. In some embodiments, each C in the antisense oligonucleotide is a 5-methylcytosine. In some embodiments, such modified nucleotides comprise a modified sugar, e.g., a modified ribose or deoxyribose. In some embodiments, the modified sugar is a 2'-O-methoxyethyl (2'-O-MOE) modified sugar, 2'-O-methyl modified sugar, or 2'-fluoro modified sugar. In some embodiments, the modified sugar is a 2'-O-MOE modified sugar. Oligonucleotides containing 5-methylcytosine or 2'-O-MOE modifications have been shown to exhibit decreased immune stimulation in mice (Henry S. et al., J Pharmacol. Exp Tuer. 2000 February; 292(2):468-

79). In some embodiments, the modified sugar is a locked nucleic acid (LNA).

[0029] In some embodiments, the antisense oligonucleotide provided herein are gapmers, which comprise a central gap segment consisting of a stretch of contiguous 2'-deoxyribonucleotides, positioned between two 5' and 3' wing segments consisting of modified nucleotides. The central gap segment of DNA can serve as the substrate for cellular endonuclease RNase H, recruit it to the target RNA:DNA heteroduplex and cleave the target RNA in the RNA:DNA duplex. The 5' and 3' wing segments with modified nucleotides confer increased resistance to other nuclease degradation. The gapmers can be in "X-Y-Z" format, where "X" represents the 5' wing region, "Y" represents the central gap region, and "Z" represents the 3' wing region. In some embodiments, the 5' and 3' wing regions comprise 2'-O-MOE modified nucleotides, and the central gap region comprises 2'-deoxynucleotides. In some embodiments, the 5' and 3' wing regions comprise LNA modified nucleotides, and the central gap region comprises 2'-deoxynucleotides.

[0030] In some embodiments, the MAPT antisense oligonucleotide provided herein are gapmers, wherein the first to fifth nucleotides each comprise a 2'-O-MOE modified sugar, wherein the sixth to fifteenth nucleotides each comprise a 2'-deoxynucleotide, and wherein the sixteenth to twentieth nucleotides each comprise a 2'-O-MOE modified sugar.

[0031] In some embodiments, the MAPT antisense oligonucleotide provided herein are gapmers, wherein the first to third nucleotides each comprise LNA, wherein the fourth to thirteenth nucleotides each comprise a 2'-deoxynucleoside, and wherein the fourteenth to sixteenth nucleotides each comprise LNA.

[0032] In some embodiments, the antisense oligonucleotides provided herein comprise one or more modified internucleotide linkages. In some embodiments, the modified internucleotide linkage is a phosphorothioate linkage. In some embodiments, the antisense oligonucleotides provided herein comprise phosphodiester and phosphorothioate internucleotide linkage. In some embodiments, the internucleotide linkages of the MAPT antisense oligonucleotides are sooooooooooooo, or sooooooooooooo, from 5' end to 3' end, wherein each s is a phosphorothioate linkage and each o is a phosphodiester linkage. In some embodiments, the internucleotide linkages of the MAPT antisense oligonucleotides are all phosphorothioate linkages.

[0033] In some embodiments, the antisense oligonucleotides provided herein comprise a sequence selected from any one of SEQ ID NOs: 12-22, 33-35, or 64-94. In some embodiments, the antisense oligonucleotides provided herein consist of a sequence selected from any one of SEQ ID NOs: 12-22, 33-35, or 64-94.

[0034] Antisense oligonucleotide can be synthesized using any nucleic acid polymerization methods known in the art, for example, solid-phase synthesis by employing phosphoramidite chemistry methodology (e.g., Current Protocols in Nucleic Acid Chemistry, Beaucage, S. L. et al. (Eds.), John Wiley & Sons, Inc., New York, NY, USA), H-phosphonate, phosphotriester chemistry, or enzymatic synthesis. Automated commercial synthesizers can be used, for example, MerMade™ 12 from LGC Biosearch Technologies, or other synthesizers from BioAutomation or Applied Biosystems. Phosphorothioate linkages can be introduced using a sulfurizing reagent such as phenylacetyl disulfide or DDTT (((dimethylaminomethylidene) amino)-3H-1,2,4-dithiazaoline-3-thione). It is well known to use similar techniques and commercially available modified amidites and controlled-pore glass (CPG) products to synthesize modified oligonucleotides or conjugated oligonucleotides.

[0035] Purification methods can be used to exclude the unwanted impurities from the final oligonucleotide product. Commonly used purification techniques for single stranded oligonucleotides include reverse-phase ion pair high performance liquid chromatography (RP-IP-HPLC), capillary gel electrophoresis (CGE), anion exchange HPLC (AX-HPLC), and size exclusion chromatography (SEC). After purification, oligonucleotides can be analyzed by mass spectrometry and quantified by spectrophotometry at a wavelength of 260 nm.

[0036] The activity of antisense oligonucleotides can be tested in vitro and/or in vivo. For in vitro

testing, the antisense oligonucleotides can be introduced into cultured cells by natural uptake, transfection or electroporation. Following a period of treatment, MAPT (tau) expression level in the ASO-treated cells can be determined and compared to MAPT (tau) expression level in the untreated control cells. MAPT expression level can be determined by any appropriate method, for example, by quantitation of MAPT mRNA level, by measuring the quantity of cDNA produced from reverse transcription of MAPT mRNA, or by determining the quantity of tau protein. These methods can be performed on a sample-by-sample basis or modified for high throughput analysis. For example, MAPT mRNA level can also be detected and quantitated by polymerase chain reaction (PCR), using a pair of primers that recognize MAPT transcript. General procedures for PCR are taught in MacPherson et al., PCR: A Practical Approach, (IRL Press at Oxford University Press (1991)). Tau protein level can be quantitated by Western blot analysis (immunoblotting), enzyme-linked immunosorbent assay (ELISA), immunohistochemistry, immunoassays, immunoprecipitation, immunofluorescent assays, immunocytochemistry, fluorescence-activated cell sorting (FACS), radioimmunoassays, immunoradiometric assays, high-performance liquid chromatography (HPLC), mass spectrometry, confocal microscopy, enzymatic assays, or surface plasmon resonance (SPR).

[0037] In another aspect, provided herein are pharmaceutical compositions comprising a MAPT antisense oligonucleotide described herein and a pharmaceutically acceptable carrier. Also provided herein are pharmaceutical compositions comprising a means for reducing MAPT expression in a cell and a pharmaceutically acceptable carrier. Such pharmaceutical compositions can also comprise one or more pharmaceutically acceptable excipient, diluent, or carrier. Pharmaceutical compositions can be prepared by methods well known in the art (e.g., Remington: The Science and Practice of Pharmacy, 23rd edition (2020), A. Loyd et al., Academic Press).

[0038] In a further aspect, provided herein are methods of reducing MAPT expression in a cell (e.g., a neuron), such methods can include contacting the cell with a MAPT antisense oligonucleotide described herein; and incubating the cell for a time sufficient for degradation of MAPT mRNA, thereby reducing MAPT expression in the cell.

[0039] In another aspect, provided herein are methods of reducing MAPT expression in a patient in need thereof, and such method comprises administering to the patient an effective amount of a MAPT antisense oligonucleotide or a pharmaceutical composition described herein. Aggregation of MAPT can be caused by overexpression of the MAPT protein or a mutation that affects the structure of the protein, resulting in an increased tendency of the MAPT protein to self-associate. Therefore, reducing MAPT expression level can be beneficial to the patient with tauopathy.

[0040] In another aspect, provided herein are methods of treating a tauopathy in a patient in need thereof, and such method comprises administering to the patient an effective amount of the MAPT antisense oligonucleotide or a pharmaceutical composition described herein. Exemplary tauopathy includes, but are not limited to, Alzheimer's disease (AD), frontotemporal dementia (FTD), frontotemporal dementia with parkinsonism linked to chromosome 17 (FTDP-17), frontotemporal lobar degeneration (FTLD), behavioral variant frontotemporal dementia (bvFTD), nonfluent variant primary progressive aphasia (nfvPPA), Parkinson's disease, Pick's disease (PiD), primary progressive aphasia-semantic (PPA-S), primary progressive aphasia-logopenic (PPA-L), multiple system tauopathy with presenile dementia (MSTD), neurofibrillary tangle (NFT) dementia, FTD with motor neuron disease, progressive supranuclear palsy (PSP), amyotrophic lateral sclerosis/parkinsonism-dementia complex (ALS-PDC), argyrophilic grain dementia (AGD), British type amyloid angiopathy, cerebral amyloid angiopathy, chronic traumatic encephalopathy (CTE), corticobasal degeneration (CBD), Creutzfeldt-Jakob disease (CJD), dementia pugilistica, diffuse neurofibrillary tangles with calcification, Down's syndrome, epilepsy, Gerstmann-Straussler-Scheinker disease, Hallervorden-Spatz disease, Huntington's disease, inclusion body myositis, lead encephalopathy, Lytico-Bodig disease (Parkinson-dementia complex of Guam), meningioangiomas, multiple system atrophy, myotonic dystrophy, Niemann-Pick disease type C

(NP-C), non-Guamanian motor neuron disease with neurofibrillary tangles, postencephalitic parkinsonism, prion protein cerebral amyloid angiopathy, progressive subcortical gliosis, tangle only dementia, tangle-predominant dementia, ganglioglioma, gangliocytoma, subacute sclerosingpan encephalitis, tuberous sclerosis, lipofuscinosis, primary age-related tauopathy (PART), globular glial tauopathies (GGT). In some embodiments, the tauopathy is Alzheimer's disease (AD), frontotemporal dementia (FTD), or progressive supranuclear palsy (PSP).

[0041] The MAPT antisense oligonucleotide can be administered to the patient intrathecally, intravenously, subcutaneously, or via intracisternal magna injection. In some embodiments, the MAPT antisense oligonucleotide is administered to the patient intrathecally.

[0042] Dosage regimens of antisense oligonucleotide may be adjusted to provide the optimum desired response (e.g., a therapeutic response). For example, a single bolus may be administered, several divided doses may be administered over time, or the dose may be proportionally reduced or increased as indicated by the exigencies of the therapeutic situation.

[0043] Dosage values may vary with the type and severity of the condition to be alleviated. It is further understood that for any particular subject, specific dosage regimens should be adjusted over time according to the individual need and the professional judgment of the person administering or supervising the administration of the compositions.

[0044] In another aspect, provided herein are MAPT antisense oligonucleotides or pharmaceutical compositions comprising a MAPT antisense oligonucleotide for use in reducing MAPT expression. Also provided herein are MAPT antisense oligonucleotides or the pharmaceutical composition comprising a MAPT antisense oligonucleotide for use in a therapy. Also provided herein are MAPT antisense oligonucleotides or pharmaceutical compositions comprising a MAPT antisense oligonucleotide for use in the treatment of a tauopathy. Also provided herein are uses of MAPT antisense oligonucleotides in the manufacture of a medicament for the treatment of a tauopathy.

[0045] As used herein, the terms “a,” “an,” “the,” and similar terms used in the context of the present disclosure (especially in the context of the claims) are to be construed to cover both the singular and plural unless otherwise indicated herein or clearly contradicted by the context.

[0046] The term “antisense oligonucleotide” as used herein refers to an oligonucleotide comprising a nucleobase sequence that is complementary to a corresponding segment of a target nucleic acid, e.g., a target genomic sequence, pre-mRNA, or mRNA molecule.

[0047] As used herein, “complementary” or “complementarity” means a structural relationship between two nucleotides, nucleosides, or nucleobases (e.g., on two opposing nucleic acids or on opposing regions of a single nucleic acid strand e.g., a hairpin) that permits the two nucleotides to form base pairs with one another. For example, a purine nucleotide of one nucleic acid that is complementary to a pyrimidine nucleotide of an opposing nucleic acid may base pair together by forming hydrogen bonds with one another. Complementary nucleotides can base pair in the canonical Watson-Crick manner, which means adenine pairing with thymine or uracil, and guanine pairing with cytosine, or in any other manner that allows for the formation of stable duplexes. Likewise, two nucleic acids may have regions of multiple nucleotides that are complementary with each other to form regions of complementarity.

[0048] An “effective amount” refers to an amount necessary (for periods of time and for the means of administration) to achieve the desired therapeutic result. An effective amount of an antisense oligonucleotide may vary according to factors such as the disease state, age, sex, and weight of the individual, and the ability of the antisense oligonucleotide to elicit a desired response in the individual. An effective amount is also one in which any toxic or detrimental effects of the antisense oligonucleotide are outweighed by the therapeutically beneficial effects.

[0049] The term “knockdown” or “expression knockdown” refers to reduced mRNA or protein expression of a gene after treatment of a reagent, e.g., an antisense oligonucleotide.

[0050] As used herein, “modified internucleotide linkage” means an internucleotide linkage having one or more chemical modifications when compared with a reference internucleotide linkage

having a phosphodiester bond. A modified internucleotide linkage can be a non-naturally occurring linkage.

[0051] As used herein, “modified nucleotide” refers to a nucleotide having one or more chemical modifications when compared with a corresponding reference nucleotide selected from: adenine ribonucleotide, guanine ribonucleotide, cytosine ribonucleotide, uracil ribonucleotide, adenine deoxyribonucleotide, guanine deoxyribonucleotide, cytosine deoxyribonucleotide, and thymidine deoxyribonucleotide. A modified nucleotide can be a non-naturally occurring nucleotide. A modified nucleotide can have, for example, one or more chemical modification in its sugar, nucleobase, and/or phosphate group. Additionally, or alternatively, a modified nucleotide can have one or more chemical moieties conjugated to a corresponding reference nucleotide.

[0052] As used herein, “nucleotide” means an organic compound having a nucleoside (a nucleobase, e.g., adenine, cytosine, guanine, thymine, or uracil, and a pentose sugar, e.g., ribose or 2'-deoxyribose) linked to a phosphate group, which can serve as a monomeric unit of nucleic acid polymers such as deoxyribonucleic acid (DNA) and ribonucleic acid (RNA).

[0053] As used herein, “oligonucleotide” means a polymer of linked nucleotides, each of which can be modified or unmodified. An oligonucleotide is typically less than about 100 nucleotides in length.

[0054] The term “patient”, as used herein, refers to a human patient.

[0055] The term “% sequence identity” or “percentage sequence identity” with respect to a reference nucleic acid sequence is defined as the percentage of nucleotides, nucleosides, or nucleobases in a candidate sequence that are identical with the nucleotides, nucleosides, or nucleobases in the reference nucleic acid sequence, after optimally aligning the sequences and introducing gaps or overhangs, if necessary, to achieve the maximum percent sequence identity. Alignment for purposes of determining percent nucleic acid sequence identity can be achieved in various ways that are within the skill in the art, for instance, using publicly available computer software programs, for example, those described in Current Protocols in Molecular Biology (Ausubel et al., eds., 1987, Supp. 30, section 7.7.18, Table 7.7.1), and including BLAST, BLAST-2, ALIGN, Clustal W2.0, Clustal X2.0, or Megalign (DNASTAR) software. Those skilled in the art can determine appropriate parameters for measuring alignment, including any algorithms needed to achieve maximal alignment over the full length of the sequences being compared. Percentage of “sequence identity” can be determined by comparing two optimally aligned sequences over a comparison window, where the fragment of the nucleic acid sequence in the comparison window may comprise additions or deletions (e.g., gaps or overhangs) as compared to the reference sequence (which does not comprise additions or deletions) for optimal alignment of the two sequences. The percentage can be calculated by determining the number of positions at which the identical nucleotide, nucleoside, or nucleobase occurs in both sequences to yield the number of matched positions, dividing the number of matched positions by the total number of positions in the window of comparison, and multiplying the result by 100 to yield the percentage of sequence identity. The output is the percent identity of the subject sequence with respect to the query sequence.

[0056] As used herein, the term “tauopathy” refers to a disease associated with abnormal tau protein expression, secretion, phosphorylation, cleavage, and/or aggregation.

[0057] As used herein, “treatment” or “treating” refers to all processes wherein there may be a slowing, controlling, delaying, or stopping of the progression of the disorders or disease disclosed herein, or ameliorating disorder or disease symptoms, but does not necessarily indicate a total elimination of all disorder or disease symptoms. Treatment includes administration of a protein or nucleic acid or vector or composition for treatment of a disease or condition in a patient, particularly in a human.

[0058] As used herein, “MAPT” refers to a human MAPT pre-mRNA or mRNA transcript, encoding a microtubule associated protein Tau. The human MAPT gene is located at chromosomal

location 17q21.1, and the genomic sequence of human MAPT gene can be found in GenBank at NG_007398.2 (SEQ ID NO: 23). The nucleotide sequences of human MAPT transcript variants and amino acid sequences of human Tau protein isoforms can be found at: [0059] i. MAPT transcript variant 1.fwdarw.Tau protein isoform 1: NM_016835.5 (nucleotide sequence).fwdarw.NP_058519.3 (amino acid sequence); [0060] ii. MAPT transcript variant 2.fwdarw.Tau protein isoform 2: NM_005910.6 (nucleotide sequence).fwdarw.NP_005901.2 (amino acid sequence); [0061] iii. MAPT transcript variant 3.fwdarw.Tau protein isoform 3: NM_016834.5 (nucleotide sequence).fwdarw.NP_058518.1 (amino acid sequence); [0062] iv. MAPT transcript variant 4.fwdarw.Tau protein isoform 4: NM_016841.5 (nucleotide sequence).fwdarw.NP_058525.1 (amino acid sequence); [0063] v. MAPT transcript variant 5.fwdarw.Tau protein isoform 5: NM_001123067.4 (nucleotide sequence).fwdarw.NP_001116539.1 (amino acid sequence); [0064] vi. MAPT transcript variant 6.fwdarw.Tau protein isoform 6: NM_001123066.4 (nucleotide sequence).fwdarw.NP_001116538.2 (amino acid sequence); [0065] vii. MAPT transcript variant 7.fwdarw.Tau protein isoform 7: NM_001203251.2 (nucleotide sequence).fwdarw.NP_001190180.1 (amino acid sequence); [0066] viii. MAPT transcript variant 8.fwdarw.Tau protein isoform 8: NM_001203252.2 (nucleotide sequence).fwdarw.NP_001190181.1 (amino acid sequence); [0067] ix. MAPT transcript variant 9.fwdarw.Tau protein isoform 9: NM_001377265.1 (nucleotide sequence).fwdarw.NP_001364194.1 (amino acid sequence); [0068] x. MAPT transcript variant 10.fwdarw.Tau protein isoform 10: NM_001377266.1 (nucleotide sequence).fwdarw.NP_001364195.1 (amino acid sequence); [0069] xi. MAPT transcript variant 11.fwdarw.Tau protein isoform 11: NM_001377267.1 (nucleotide sequence).fwdarw.NP_001364196.1 (amino acid sequence); [0070] xii. MAPT transcript variant 12.fwdarw.Tau protein isoform 4: NM_001377268.1 (nucleotide sequence).fwdarw.NP_001364197.1 (amino acid sequence).

[0071] The nucleotide sequence of the human MAPT transcript variant 6 (encoding 2N4R Tau) can be found at NM_001123066.4 (SEQ ID NO: 24). The corresponding amino acid sequence of human Tau protein isoform 6 can be found at NP_001116538.2 (SEQ ID NO: 25).

[0072] The nucleotide sequence of a human MAPT transcript variant 5 (encoding 1N4R Tau) can be found at NM_001123067.4 (SEQ ID NO: 26). The corresponding amino acid sequence of human Tau protein isoform 5 can be found at NP_001116539.1 (SEQ ID NO: 27).

[0073] The nucleotide sequence of the human MAPT transcript variant 4 (encoding 0N3R Tau) can be found at NM_016841.5 (SEQ ID NO: 28). The corresponding amino acid sequence of human Tau protein isoform 4 can be found at NP_058525.1 (SEQ ID NO: 29).

EXAMPLES

Example 1. Synthesis of MAPT Antisense Oligonucleotides

[0074] The oligonucleotide sequences shown in Table 1 were synthesized on solid support via phosphoramidite chemistry using the Shasta Plate Synthesizer (Sierra Biosystems) for 100 nmol scale syntheses and the H8-SE Synthesizer (K&A) for 10 μ mol scale syntheses.

[0075] Standard reagents were used in the oligonucleotide synthesis (Table 2), where 0.1M DDTT in pyridine was used as the sulfurization reagent. All phosphoramidites were commercially available (ChemGenes) and were prepared at 0.1M in Acetonitrile and contained a molecular sieves trap bag.

[0076] Oligonucleotides synthesized at the 0.1 nmol scale on 96-well plates were cleaved and deprotected (C/D) using ammonium hydroxide in the gas phase at 80° C. and at 40 psi. Wells were washed twice with 100 μ L of anhydrous acetonitrile and then dried. Oligonucleotides were eluted in 300 μ L of water, analyzed for concentration using absorbance at 260 nm recorded by the Lunatic (Unchained Labs), and characterized by LC-QTOF for mass purity of >85%.

[0077] Oligonucleotides synthesized at the 10 μ mol scale were cleaved and deprotected (C/D)

using a 1:1 solution of 25% aqueous ammonium hydroxide and 40% aqueous methylamine at room temperature for 2 hours. The solution containing cleaved oligonucleotide was decanted and concentrated in the Genevac for the ammonia and methylamine removal. The CPG was filtered via 0.45 μ m PVDF filter.

[0078] The crude oligonucleotides were purified via AKTA™ Pure purification system using anion-exchange (AEX) with a HiScale 16 Capto Q ImpRes column (Cytiva Life Sciences) with MPA: 20 mM NaOH, 20% ACN, pH 12.4 and MPB: 20 mM NaOH, 1.5M NaBr, 20% ACN, pH 12.4.

[0079] The purified oligonucleotides were desalted in 15 mL 3K MWCO centrifugal spin tubes by rinsing with RNase free water at 3500×g for ~30 min. The final desalted oligonucleotides were analyzed for concentration (nano drop at A260), characterized by LC-QTOF for mass purity and UPLC for UV-purity.

TABLE-US-00002 TABLE 2 Oligonucleotide Synthesis Reagents Reagents Activator Solution (0.45M ETT in ACN) Cap A (Acetic Anhydride, Pyridine in THF, 1:1:8) Cap B (1-Methylimidazole in THF, 16:84) Oxidation Solution (0.02M Iodine in THF/Pyridine/Water, 70:20:10) Deblock Solution, 3% TCA in DCM (w/v) Acetonitrile (Anhydrosolv, Water max. 10 ppm) DDTT (0.1M in Pyridine)

Example 2. In Vitro Characterization of MAPT Antisense Oligonucleotides

[0080] Selected MAPT antisense oligonucleotides were tested in vitro for MAPT inhibition.

Materials and Methods

[0081] SH-SY5Y Cell Culture and MAPT Antisense Oligonucleotide Treatment: SH-SY5Y cells (ATCC CRL-2266) were derived from the SK-N-SH neuroblastoma cell line (Ross, R. A., et al., 1983. J Natl Cancer Inst 71, 741-747). The base medium was composed of a 1:1 mixture of ATCC-formulated Eagle's Minimum Essential Medium, (Cat No. 30-2003), and F12 Medium. The complete growth medium was supplemented with 10% fetal bovine serum, and cells were incubated at 37° C. in a humidified atmosphere of 5% CO₂. On Day One, antisense oligonucleotide and Lipofectamine RNAiMax transfection reagent (ThermoFisher 13778030) complexes were prepared inside the wells and SH-SY5Y cells were plated in 96 well tissue culture plates and incubated for 24 hrs. On Day Two, analysis of changes in gene expression in the treated SH-SY5Y cells was measured using TaqMan™ Gene Expression Cells-to-CT™ Kits following the manufacturer's protocol (ThermoFisher AM1729). Predesigned gene expression assays (supplied as 60× or 20× mixtures) were selected from TaqMan Real-Time PCR Assays. The efficiencies of these assays for MAPT (ThermoFisher Hs00902193_m1) and TBP (ThermoFisher #4326322E) were characterized with a dilution series of cDNA. RT-QPCR was performed in MicroAmp Optical 384-well reaction plates using QuantStudio 7 Pro system. The delta-delta CT method of normalizing to the transfection only control sample was used to determine relative amounts of MAPT gene expression. GraphPad Prism v10.2 was used to determine absolute or relative IC₅₀ with a four-parameter logistic fit.

[0082] Human induced pluripotent stem cells (hiPSCs) derived glutamatergic neurons (iCell GlutanNeurons) (CDI, 01279) was obtained from Fujifilm Cellular Dynamics International (CDI). iCell GlutanNeurons were handled and maintained following CDI's guidelines. Tissue culture treated 96 well plates (Thermo Scientific, 167574) were coated with poly-L-ornithine (PLO) and Matrigel (Corning 354230). 80,000 cells were plated in each well and were grown in Complete BrainPhys Medium consisting of BrainPhys Neuronal Medium (STEMCELL Technologies, 05790), iCell Neural Supplement B (CDI, M1029), iCell Nervous System Supplement (CDI, M1031), N-2 Supplement (Thermo Scientific, 17502-048) and Laminin (Sigma-Aldrich, L2020) in an incubator (37°/5% CO₂). Starting the day after plating, half of media was replenished every other day for seven days. On the eighth day, antisense oligonucleotide was serially diluted in Complete BrainPhys Medium, and cells were treated with MAPT antisense oligonucleotide by aspirating half of media from each well (100 μ L) and adding 100 μ L of 2×MAPT antisense

oligonucleotide concentration for a final of 1×MAPT antisense oligonucleotide according to dilutions. Cells were half-fed either every three or four days after treatment by removing half of media and adding back fresh Complete BrainPhys Medium with 2×MAPT antisense oligonucleotide concentration. Cell lysates were harvested after seven days of treatment (14 days post seeding) and RT-qPCR was performed using TaqMan Fast Advanced Cells-to CT Kit (Thermo Scientific, A35377) to determine mRNA knock down level. MAPT probe (ThermoFisher, Hs00902193_m1) was used as the gene of interest while TBP probe was used as the reference gene (ThermoFisher, 4326322E). RT-QPCR was performed in MicroAmp Optical 384-well reaction plates using the QuantStudio 7 Pro system. The delta-delta CT method was employed to normalize to the non-treated control sample and determine relative MAPT gene expression. GraphPad Prism v10.2 was used to calculate or relative IC.sub.50 value with a four-parameter logistic fit.

[0083] Table 3 summarizes the in vitro activities of selected MAPT antisense oligonucleotides in SH-SY5Y cells. As shown in Table 3, the tested antisense oligonucleotides knock down MAPT expression in vitro in SH-SY5Y cells.

TABLE-US-00003 TABLE 3 In vitro activities of selected MAPT antisense oligonucleotides in SH-SY5Y cells														
SH-SY5Y 1 d	SH-SY5Y 1 d	Absolute IC.sub.50	% KD (knockdown)	MAPT ASO (nM)	mRNA at 10 nM	No Transfection	Transfection	1	0.41	93.9 ± 0.4	2	2.06	90.9 ± 0.6	3
0.73	90.3 ± 2.4	4	1.69	89.5 ± 2.5	5	0.37	89.1 ± 1.8	6	1.23	89.1 ± 1.7	7	0.74	87.5 ± 1.9	8
21.95	86.1 ± 3.9	9	1.94	84.8 ± 1.9	10	4.90	84.1 ± 2.1	11	0.70	84.1 ± 1.9	12	0.25	77.41 ± 2.1	13
1.55	82.4 ± 3.3	14	10.38	83.2 ± 0.5										

[0084] Table 4 summarizes the in vitro activities of selected MAPT antisense oligonucleotides in human iPSC cells. As shown in Table 4, the tested antisense oligonucleotides knock down MAPT expression in human iPSC cells.

TABLE-US-00004 TABLE 4 In vitro activities of selected MAPT antisense oligonucleotides in human iPSC cells														
iPSC 7 d	iPSC 7 d	MAPT ASO	Relative IC.sub.50	% KD (knockdown)	MAPT No (μM)	mRNA at 10 nM	15	2.916	70.11	16	10.4	44.69	17	3.562
54.88	18	10.66	26.92	19	9.743	33.38	20	1.782	78.32	21	2.782	64.02	22	7.396
43.72	23	1.08	80.31	24	0.6954	81.3	25	2.214	64.58	26	0.956	79.54	27	0.05902
98	28	ND*	94.83	29	ND*	85.99	30	ND*	96.75	31	ND*	61.96	32	0.057
97.35	33	0.018	99.37	34	0.016	98.41	35	ND*	97.04	36	ND*	90.86	37	0.012
98.39	38	ND*	68.99	39	ND*	56.67	40	ND*	58.94	41	ND*	34.01	42	ND*
67.05	43	6.04	53.89	44	6.64	57.38	45	2.19	70.57	ND*				

ND* indicates not determined.

Example 3. In Vivo Characterization of MAPT Antisense Oligonucleotides

[0085] Selected MAPT antisense oligonucleotides were tested in vivo for MAPT inhibition in human MAPT transgenic mice.

Materials and Methods

[0086] Mouse Stereotaxic Intracerebroventricular (ICV) Surgery and MAPT Antisense Oligonucleotide Treatment: C57/BL6 transgenic mice expressing human MAPT were placed in an induction chamber (1-5% Isoflurane using a calibrated vaporizer) after administering 5 mg/kg of ketoprofen (subcutaneous). Hair is removed from the surgical site using electric clippers. The surgical area is disinfected with iodine and alcohol three times to create an aseptic surgical field. Intracerebroventricular (ICV) injection was performed, and the animal is returned to its cage. 6 mice per group received ICV injection of 150 μg of the MAPT ASO or PBS (phosphate buffered saline).

[0087] After a 14-day incubation, animals were euthanized. The tissues of interest were harvested and weighed. The tissues are then stored in RNeasy[™] stabilization solution (Invitrogen[™] catalog number AM7024). Once all the tissues were collected, RNA was extracted from the tissues using the protocol for the RNeasy Kit (Qiagen[™] catalog number. 74182. Reverse transcription was then performed using the High-Capacity cDNA RT Master mix (Applied Biosystems[™] catalog number. 4374967). Analysis of changes in hMAPT expression from the tissue homogenates was measured using Fast-Advanced TaqMan[™] Master Mix (Applied Biosystems[™] catalog number. 4444557)

along with validated pre-designed TaqMan primers for human MAPT (Hs00902193_m1) and mouse GAPDH (Mm99999915_g1) as a housekeeping gene. The qPCR was performed in MicroAmp Optical 384-well reaction plates using QuantStudio 7 Pro system. The delta-delta CT method of normalizing against samples that were treated with PBS control was used to determine relative amounts of MAPT gene expression.

TABLE-US-00005 TABLE 5 summarizes the in-vivo activities of selected MAPT antisense oligonucleotides in different tissues of transgenic mice expressing the human MAPT gene.

Antisense MAPT	Frontal Brain	Spinal ASO	No Cortex	Striatum	Cerebellum	Stem Cord	PBS
100.0	100.0	100.0	100.0	100.0	7.825	8.807	5.194
4.056	3.726	1	40.53	55.08	102.3	57.91	79.44
8.643	6.910	6.262	9.432	7.965	2	49.14	65.28
100.3	66.55	91.97	4.246	5.187	3.412	4.031	1.882
3	58.81	66.55	91.99	67.61	89.58	4.871	6.022
4.492	5.137	5.040	4	56.30	63.68	88.19	65.54
95.85	8.123	7.089	4.695	4.362	9.168	5	50.69
46.86	99.39	48.22	98.84	8.700	4.532	4.039	4.727
6.570	6	56.41	52.97	101.2	44.13	94.31	5.546
3.938	4.742	3.688	2.996	7	95.84	95.15	110.8
137.1	96.31	6.839	10.50	6.958	10.63	3.672	8
69.89	74.51	113.2	134.6	101.7	6.987	10.57	4.399
11.97	4.444	9	69.67	78.68	124.8	97.14	96.24
6.117	6.420	7.841	13.59	6.407	10	76.27	78.33
115.9	91.91	103.6	9.847	4.310	2.247	8.329	6.060
11	62.31	61.20	97.20	23.91	71.36	7.683	4.366
6.065	2.600	9.038					

Sequence Listing

TABLE-US-00006 SEQ ID NO Sequence 1 CTTTCTTGGAAGTATATTC 2
GCTCAGAGCTACAGCAAGTC 3 GCTGCAGGGAAAGAGTTCTT 4
GATGGAGACAGTGTGTATCT 5 GCAGATGGGGCCCAAATCCT 6
GCCACAGTGACAGCAAATTC 7 GCCACAGACTCGAGTATCAG 8
CCAAACGTGACTTGGCTCCT 9 TATGGACACACATGGGTGTA 10
ACTAAAGACGTGTCTTTTGG 11 GTCATCTCTAGGTGACTTCC 12
[eCm]seToeToeTo[eCm]sdTsdTsdGsdGsdAsdAsdAsdGsdTsdAseToeAoeTseTs[eCm] 13
eGs[eCm]oeTo[eCm]oeAsdGsdAsdGs[dCm]sdTsdAs[dCm]sdAsdGs[dCm]seAoeAoeGseTs[eCm] 14
eGs[eCm]oeToeGo[eCm]sdAsdGsdGsdGsdAsdAsdAsdGsdAsdGseToeTo[eCm]seTseT 15
eGseAoeToeGoeGsdAsdGsdAs[dCm]sdAsdGsdTsdGsdTsdGseToeAoeTs[eCm]seT 16
eGs[eCm]oeAoeGoeAsdTsdGsdGsdGsdGs[dCm]s[dCm]s[dCm]sdAsdAseAoeTo[eCm]s[eCm]seT 17
eGs[eCm]o[eCm]oeAo[eCm]sdAsdGsdTsdGsdAs[dCm]sdAsdGs[dCm]sdAseAoeAoeTseTs[eCm] 18
eGs[eCm]o[eCm]oeAo[eCm]sdAsdGsdAs[dCm]sdTs[dCm]sdGsdAsdGsdTseAoeTo[eCm]seAseG 19
[eCm]s[eCm]oeAoeAoeAs[dCm]sdGsdTsdGsdAs[dCm]sdTsdTsdGsdGs[eCm]oeTo[eCm]s[eCm]seT 20
eTseAoeToeGoeGsdAs[dCm]sdAs[dCm]sdAs[dCm]sdAsdTsdGsdGseGoeToeGseTseA 21
eAs[eCm]oeToeAoeAsdAsdGsdAs[dCm]sdGsdTsdGsdTs[dCm]sdTseToeToeTseGseG 22
eGseTo[eCm]oeAoeTs[dCm]sdTs[dCm]sdTsdAsdGsdGsdTsdGsdAs[eCm]oeToeTs[eCm]s[eCm] 23

1	GCAGTCACCG	CCACCCACCA	GCTCCGGCAC	CAACAGCAGC
GCCGCTGCCA	CCGCCCACCT	61	TCTGCCGCCG	CCACCACAGC
CACCTTCTCC	TCCTCCGCTG	TCCTCTCCCG	TCCTCGCCTC	121
TGTCGACTAT	CAGGTAAGCG	CCGCGGCTCC	GAAATCTGCC	TCGCCGTCCG
CCTCTGTGCA	181	CCCCTGCGCC	GCCGCCCTC	GCCCTCCCTC
TCCGCAGACT	GGGGCTTCGT	GCGCCGGGCA	241	TCGGTCGGGG
CCACCGCAGG	GCCCCTCCCT	GCCTCCCCTG	CTCGGGGGCT	GGGGCCAGGG
301	CGGCCTGGAA	AGGGACCTGA	GCAAGGGATG	CACGCACGCG
TGAGTGCGCG	CGTGTGTGTG	361	TGCTGGAGGG	TCTTCACCAC
CAGATTCGCG	CAGACCCAG	GTGGAGGCTG	TGCCGGCAGG	421
GTGGGGCGCG	GCGGCGGTGA	CTTGGGGGAG	GGGGCTGCC	TTCACTCTCG

ACTGACGACCT 481 TTGCCGCAA TGGCGTGTG TGTGTGTG
 TGTGTGTGTG TGTGTGTGTG TGTGTGTGGA 541 GGGGTCCGAT
 AACGACCCCC GAAACCGAAT CTGAAATCCG CTGTCCCTGC CGCTGTTCGC
 601 CATCAGCTCT AAGAAAGACG TGGATCGGGT TCTAGAAAAG
 ATGACTCCCT GCACGCCCCT 661 CCCTGCACCT CCCGAGCAGT
 GATTCCGACA GGGCCTTCAC TGCCCTGAT TTTAGGCGGG 721
 GGCCGGCCCC CTCCCCTTTT CCTCCTTCAG AAACCCGTAG GGGACATTG
 GGGGCTGGGA 781 GAAATCGAGG AGATGGGGAG GGGTCCACGC
 GCTGTCACTT TAGTTGCCCT TCCCCCTGCG 841 CACGCCTGGC
 ACAGAGACGC GAGCAGCGCC GTGCCTGAGA ACAGTGCGCG GATCCCACTG
 901 TGCACGCTCG CAAAGGCAGG GTTCACCTGG CCTGGCGATG
 TGGACGGACT CGGCGGCCGC 961 TGGTCCCCGT TCGCGGGCAC
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Claims

1. A MAPT antisense oligonucleotide comprises a nucleic acid sequence selected from any one of SEQ ID NOs: 1-11, 30-32, or 36-63, wherein optionally one or more nucleotides are independently

- modified nucleotides, and wherein optionally one or more internucleotide linkages are modified internucleotide linkages.
2. The MAPT antisense oligonucleotide of claim 1, wherein the antisense oligonucleotide is single stranded.
 3. The MAPT antisense oligonucleotide of claim 1, wherein the antisense oligonucleotide comprises one or more modified nucleotides.
 4. The MAPT antisense oligonucleotide of claim 1, wherein the one or more modified nucleotides comprise a modified nucleobase.
 5. The MAPT antisense oligonucleotide of claim 4, wherein the modified nucleobase is a 5-methylcytosine.
 6. The MAPT antisense oligonucleotide of claim 5, wherein each C in the antisense oligonucleotide is a 5-methylcytosine.
 7. The MAPT antisense oligonucleotide of claim 1, wherein the one or more modified nucleotides comprise a modified sugar.
 8. The MAPT antisense oligonucleotide of claim 7, wherein the modified sugar is a 2'-O-methoxyethyl (2'-O-MOE) modified sugar, 2'-O-methyl modified sugar, 2'-fluoro modified sugar, or a locked nucleic acid (LNA).
 9. The MAPT antisense oligonucleotide of claim 8, wherein the modified sugar is a 2'-O-MOE modified sugar.
 10. The MAPT antisense oligonucleotide of claim 8, wherein the modified sugar is LNA.
 11. The MAPT antisense oligonucleotide of claim 1, wherein the antisense oligonucleotide is 16-20 nucleotides in length.
 12. The MAPT antisense oligonucleotide of claim 1, wherein the first to fifth nucleotides each comprise a 2'-O-MOE modified sugar, wherein the sixth to fifteenth nucleotides each comprise a 2'-deoxynucleotide, and wherein the sixteenth to twentieth nucleotides each comprise a 2'-O-MOE modified sugar.
 13. The MAPT antisense oligonucleotide of claim 1, wherein the first to third nucleotides each comprise LNA, wherein the fourth to thirteenth nucleotides each comprise a 2'-deoxynucleoside, and wherein the fourteenth to sixteenth nucleotides each comprise LNA.
 14. The MAPT antisense oligonucleotide of claim 1, wherein the antisense oligonucleotide comprises one or more modified internucleotide linkages.
 15. The MAPT antisense oligonucleotide of claim 14, wherein the one or more modified internucleotide linkages are phosphorothioate linkages.
 16. The MAPT antisense oligonucleotide of claim 14, wherein the internucleotide linkages of the antisense oligonucleotide are sooooooooooooo, or sooooooooooooooooo, from 5' end to 3' end, wherein each s is a phosphorothioate linkage and each o is a phosphodiester linkage.
 17. A MAPT antisense oligonucleotide comprises a sequence selected from any one of SEQ ID NOs: 12-22, 33-35, or 64-94.
 18. The MAPT antisense oligonucleotide of claim 17, wherein the antisense oligonucleotide consists of a sequence selected from any one of SEQ ID NOs: 12-22, 33-35, or 64-94.
 19. A pharmaceutical composition comprising the MAPT antisense oligonucleotide of claim 1 and a pharmaceutically acceptable carrier.
 20. A pharmaceutical composition comprising the MAPT antisense oligonucleotide of claim 17 and a pharmaceutically acceptable carrier.
 21. A method of reducing MAPT expression in a patient in need thereof, the method comprising administering to the patient an effective amount of the MAPT antisense oligonucleotide of claim 1.
 22. A method of reducing MAPT expression in a patient in need thereof, the method comprising administering to the patient an effective amount of the MAPT antisense oligonucleotide of claim 17.
 23. A method of treating a tauopathy in a patient in need thereof, the method comprising

administering to the patient an effective amount of the MAPT antisense oligonucleotide of claim 1.

24. The method of claim 23, wherein the tauopathy is selected from Alzheimer's disease (AD), frontotemporal dementia (FTD), frontotemporal dementia with parkinsonism linked to chromosome 17 (FTDP-17), frontotemporal lobar degeneration (FTLD), behavioral variant frontotemporal dementia (bvFTD), nonfluent variant primary progressive aphasia (nfvPPA), Parkinson's disease, Pick's disease (PiD), primary progressive aphasia-semantic (PPA-S), primary progressive aphasia-logopenic (PPA-L), multiple system tauopathy with presenile dementia (MSTD), neurofibrillary tangle (NFT) dementia, FTD with motor neuron disease, progressive supranuclear palsy (PSP), amyotrophic lateral sclerosis/parkinsonism-dementia complex (ALS-PDC), argyrophilic grain dementia (AGD), British type amyloid angiopathy, cerebral amyloid angiopathy, chronic traumatic encephalopathy (CTE), corticobasal degeneration (CBD), Creutzfeldt-Jakob disease (CJD), dementia pugilistica, diffuse neurofibrillary tangles with calcification, Down's syndrome, epilepsy, Gerstmann-Straussler-Scheinker disease, Hallervorden-Spatz disease, Huntington's disease, inclusion body myositis, lead encephalopathy, Lytico-Bodig disease, meningioangiomas, multiple system atrophy, myotonic dystrophy, Niemann-Pick disease type C (NP-C), non-Guamanian motor neuron disease with neurofibrillary tangles, postencephalitic parkinsonism, prion protein cerebral amyloid angiopathy, progressive subcortical gliosis, tangle only dementia, tangle-predominant dementia, ganglioglioma, gangliocytoma, subacute sclerosing pan encephalitis, tuberous sclerosis, lipofuscinosis, primary age-related tauopathy (PART), or globular glial tauopathies (GGT).

25. The method of claim 23, wherein the MAPT antisense oligonucleotide is administered to the patient intrathecally, intravenously, subcutaneously, or via intracisternal magna injection.

26. A method of treating a tauopathy in a patient in need thereof, the method comprising administering to the patient an effective amount of the MAPT antisense oligonucleotide of claim 17.

27. The method of claim 26, wherein the tauopathy is selected from Alzheimer's disease (AD), frontotemporal dementia (FTD), frontotemporal dementia with parkinsonism linked to chromosome 17 (FTDP-17), frontotemporal lobar degeneration (FTLD), behavioral variant frontotemporal dementia (bvFTD), nonfluent variant primary progressive aphasia (nfvPPA), Parkinson's disease, Pick's disease (PiD), primary progressive aphasia-semantic (PPA-S), primary progressive aphasia-logopenic (PPA-L), multiple system tauopathy with presenile dementia (MSTD), neurofibrillary tangle (NFT) dementia, FTD with motor neuron disease, progressive supranuclear palsy (PSP), amyotrophic lateral sclerosis/parkinsonism-dementia complex (ALS-PDC), argyrophilic grain dementia (AGD), British type amyloid angiopathy, cerebral amyloid angiopathy, chronic traumatic encephalopathy (CTE), corticobasal degeneration (CBD), Creutzfeldt-Jakob disease (CJD), dementia pugilistica, diffuse neurofibrillary tangles with calcification, Down's syndrome, epilepsy, Gerstmann-Straussler-Scheinker disease, Hallervorden-Spatz disease, Huntington's disease, inclusion body myositis, lead encephalopathy, Lytico-Bodig disease, meningioangiomas, multiple system atrophy, myotonic dystrophy, Niemann-Pick disease type C (NP-C), non-Guamanian motor neuron disease with neurofibrillary tangles, postencephalitic parkinsonism, prion protein cerebral amyloid angiopathy, progressive subcortical gliosis, tangle only dementia, tangle-predominant dementia, ganglioglioma, gangliocytoma, subacute sclerosing pan encephalitis, tuberous sclerosis, lipofuscinosis, primary age-related tauopathy (PART), or globular glial tauopathies (GGT).

28. The method of claim 26, wherein the MAPT antisense oligonucleotide is administered to the patient intrathecally, intravenously, subcutaneously, or via intracisternal magna injection.

29. A method of reducing MAPT expression in a cell, the method comprising: contacting the cell with the MAPT antisense oligonucleotide of claim 1; and incubating the cell for a time sufficient for degradation of MAPT mRNA, thereby reducing MAPT expression in the cell.

30. A method of reducing MAPT expression in a cell, the method comprising: contacting the cell

with the MAPT antisense oligonucleotide of claim 17; and incubating the cell for a time sufficient for degradation of MAPT mRNA, thereby reducing MAPT expression in the cell.
